

Global Water Quality & Health Impact Analysis

Presented By:

Archand Raj

Batch : PGDDM52

Course : Data Science And Machine Learning

Introduction

- ▶ **We live in an era of rapid environmental change where monitoring water pollution has become critical for public health and ecosystem survival.** Safe and clean water is a fundamental necessity, but human activities, industrial runoffs, and changing weather patterns constantly threaten its quality.
- ▶ **Regular monitoring of key parameters is one of the most effective ways to track and mitigate contamination before it reaches critical levels.** However, water safety is not uniform. Pollution metrics vary dynamically across different regions, seasons, and water sources based on multiple interacting environmental factors
- ▶ Understanding these variations through data-driven analysis is essential to identify the root causes of contamination, predict pollution spikes, and develop highly effective, proactive water treatment and resource management strategies.

Objective

- ▶ The main objective of this project is to analyze the key environmental metrics in the water pollution dataset and understand the factors that drive water contamination and quality degradation.
- ▶ **This includes examining how variables such as water source, region, season, temperature, and rainfall impact critical metrics like turbidity, pH level, dissolved oxygen, chemical concentrations, and overall safety index.** The goal is to identify patterns, relationships, and trends that explain variations in water quality.
- ▶ The project aims to provide data-driven insights that can help environmentalists, municipal authorities, and treatment plants make better decisions regarding water resource management, contamination control, and infrastructure planning.

Business Understanding

- ▶ Identify high-risk contamination zones and sources
- ▶ Understand the impact of environmental factors on water quality
- ▶ Optimize water treatment and mitigation resources

Data Overview

- ▶ This dataset contains **30,000 records** of water quality metrics collected from various geographical regions and water sources.
- ▶ It includes a comprehensive range of columns covering both environmental conditions and chemical safety indicators, providing a multi-dimensional view of water health.
- ▶ The dataset consists of information related to regional locations, water body types, and seasonal weather patterns, along with key quality metrics such as turbidity, pH levels, dissolved oxygen, and chemical concentrations.
- ▶ The data includes both categorical variables (e.g., region, water source) and numerical variables (e.g., rainfall, contamination levels), making it suitable for comparison, correlation, and trend analysis.
- ▶ This dataset is useful for understanding how environmental factors influence water safety and for identifying patterns in pollution spikes across different regions and seasons.

Dataset

	Country	Region	Year	Water Source Type	Contaminant Level (ppm)	pH Level	Turbidity (NTU)	Dissolved Oxygen (mg/L)	Nitrate Level (mg/L)	Lead Concentration (µg/L)	...
0	Mexico	North	2014.573011	Lake	5.949580	NaN	3.975384	4.380503	6.839763	8.212500	...
1	Brazil	West	2017.168800	Well	5.101385	7.821342	4.784853	3.842380	15.274648	15.120733	...
2	Indonesia	Central	2021.662742	Pond	0.378940	6.468505	0.918661	3.510669	36.201394	9.757347	...
3	Nigeria	East	2016.099418	Well	7.713726	6.731788	2.096610	3.245622	37.058182	7.222446	...
4	Mexico	South	2004.540148	Well	0.098594	NaN	4.274853	9.067828	49.247285	12.793763	...
...	...	...	...	...	...	...	...	...	...	...	...
29995	Nigeria	East	2002.111421	Tap	2.609618	7.435803	4.428310	9.824740	38.409297	18.374965	...
29996	Nigeria	North	2018.992895	Lake	8.009406	8.345291	4.828078	7.693366	38.568742	17.236935	...
29997	USA	South	2008.723261	Pond	1.190236	6.748994	4.753743	7.121178	35.952761	7.791153	...
29998	Pakistan	South	2008.994324	Pond	7.761946	6.112992	3.563171	8.972651	25.178188	19.757613	...
29999	Mexico	North	2013.116169	Spring	1.216115	7.649977	2.390441	7.621681	19.338006	1.160997	...

30000 rows × 24 columns

Cholera Cases per 100,000 people	Typhoid Cases per 100,000 people	Infant Mortality Rate (per 1,000 live births)	GDP per Capita (USD)	Healthcare Access Index (0-100)	Urbanization Rate (%)	Sanitation Coverage (% of Population)	Rainfall (mm per year)	Temperature (°C)	Population Density (people per km²)
33.059178	42.330220	75.394165	57965.182638	NaN	83.922388	62.272682	2771.119295	4.960966	567.343524
27.885419	8.437984	75.303631	15996.016419	88.679832	73.260644	28.900040	1468.569987	17.165776	239.024066
37.707123	52.185247	47.177769	86705.267673	56.836526	72.766013	94.497151	2111.274630	21.440669	50.862682
32.849930	14.168766	92.689431	31773.715604	39.045881	72.570736	92.487110	971.770615	3.845061	545.369767
31.000651	67.321848	61.237133	25246.898248	23.106234	55.379023	69.653695	2267.711824	31.234929	407.501548
...	...	...	...	...	...	...	...	...	...
29.874443	55.895654	73.542209	22567.882193	75.829698	12.160032	55.374876	275.994891	31.042524	893.697650
34.442142	34.085015	72.103693	77516.173732	7.955315	25.047696	78.690709	2723.201070	34.001430	45.933373
48.506122	10.961165	94.327441	49541.319926	87.436399	88.156582	46.507190	1046.637656	10.972089	648.683663
33.067969	90.567501	20.013129	51069.133054	19.732511	23.390177	24.093279	420.564793	14.729642	635.050721
3.378804	80.865251	-0.208861	70882.362803	43.801610	76.631958	67.699254	872.390894	10.276807	280.879619

Data Source

- ▶ **The dataset used in this project is a comprehensive collection of environmental records designed to simulate or represent real-world water quality monitoring scenarios.**
- ▶ It represents realistic patterns of environmental behavior, including chemical concentrations, physical water properties, and weather-related impacts.
- ▶ **The data includes information from multiple geographical locations and diverse water body types, such as rivers, lakes, and groundwater reserves.**
- ▶ This dataset is primarily used for educational and analytical purposes, helping to identify trends and risk factors in global water safety management.

EDA Process

- ❑ Data Loading
- ❑ Understanding the Data (`head()`, `info()`, `describe()`)
- ❑ Data Cleaning (missing values, duplicates, data types)
- ❑ Data Categorization (categorical vs numerical)
- ❑ Univariate Analysis (single variable analysis)
- ❑ Bivariate Analysis (relationship between two variables)
- ❑ Multivariate Analysis (multiple variable relationships)
- ❑ Data Visualization (charts and graphs)

EDA Using Packages

- ▶ **Exploratory Data Analysis (EDA)** is an essential step in the data analysis process that helps in understanding the structure, patterns, and key characteristics of a dataset. It involves summarizing data, detecting anomalies, and identifying relationships between variables. There are several Python packages commonly used for performing EDA. Some of the popular ones include:
- ▶ **Pandas:** Pandas is a powerful library used for data manipulation and analysis. It provides data structures like DataFrames and Series, which are useful for loading, cleaning, and exploring datasets efficiently.
- ▶ **NumPy:** NumPy is used for numerical computations and supports operations on large arrays and matrices. It is often used alongside Pandas for performing mathematical and statistical calculations.
- ▶ **Matplotlib:** Matplotlib is a widely used visualization library that helps in creating various types of plots such as line charts, bar charts, histograms, and scatter plots to understand data trends and distributions.
- ▶ **Seaborn:** Seaborn is built on top of Matplotlib and provides a higher-level interface for creating visually appealing and informative statistical graphics. It simplifies complex visualizations like heatmaps, box plots, and violin plots.

Data Cleaning

- ▶ **Handling Missing Values:** Identified and addressed null entries in critical columns like pH levels or chemical concentrations to maintain dataset integrity.

Check for missing values in the dataset

```
print(df.isnull().sum())
```

Filling missing numerical values (like pH or Turbidity) with the mean

```
df['pH_Level'] = df['pH_Level'].fillna(df['pH_Level'].mean())
```

Dropping rows where critical data is missing

```
df.dropna(subset=['Contaminant_Type'], inplace=True)
```

- ▶ **Removing Duplicates:** Scanned the dataset for redundant water sample records to ensure that specific pollution incidents weren't over-counted.

Check for duplicate rows

```
print(df.duplicated().sum())
```

Remove duplicate entries

```
df.drop_duplicates(inplace=True)
```

- ▶ **Standardizing Formats:** Ensured consistent naming conventions for regions and water sources.

Standardizing string formats to lowercase or title case

```
df['Water_Source'] = df['Water_Source'].str.strip().str.title()
```

- ▶ **Data Type Conversion:** Verified that numerical columns (like Turbidity and Rainfall) were correctly formatted as floats/integers for mathematical analysis.

Converting columns to numeric, forcing errors to NaN to handle "messy" strings

```
df['Turbidity_Level'] = pd.to_numeric(df['Turbidity_Level'], errors='coerce')
```

- ▶ **Outlier Detection:** Inspected extreme values in pollution metrics to determine if they were genuine contamination spikes or data entry errors.

Finding outliers using IQR

```
Q1 = df['Chemical_Concentration'].quantile(0.25)
```

```
Q3 = df['Chemical_Concentration'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

Defining bounds

```
lower_bound = Q1 - 1.5 * IQR
```

```
upper_bound = Q3 + 1.5 * IQR
```

Filtering the data

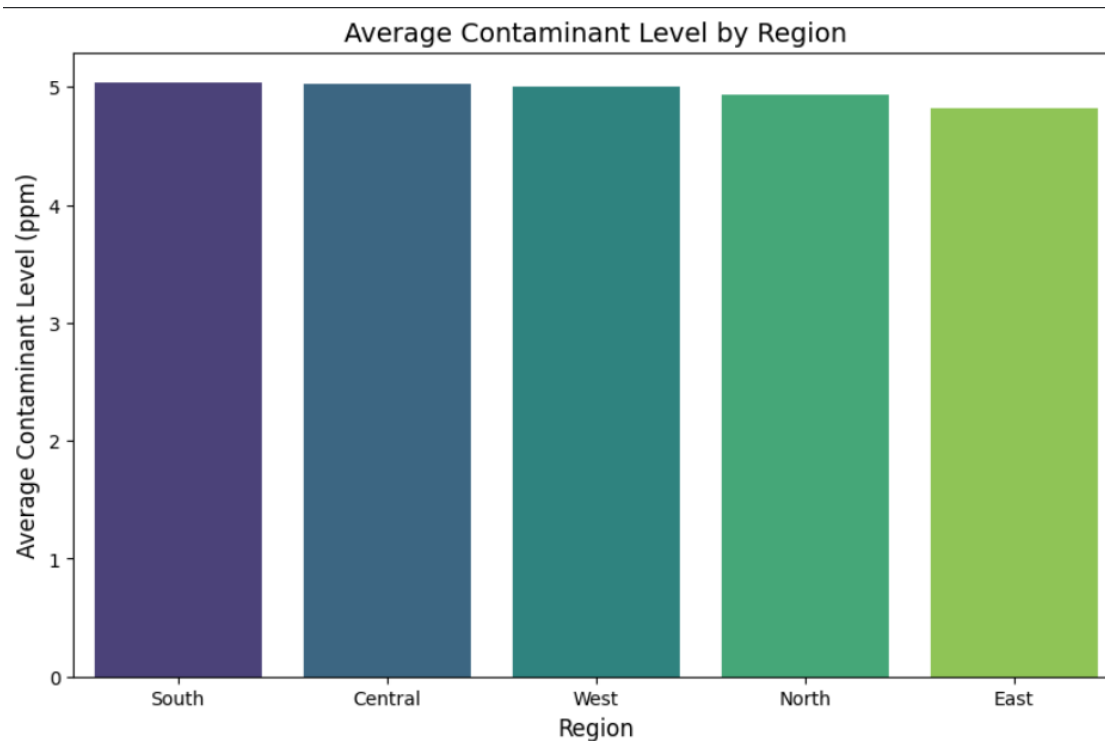
```
outliers = df[(df['Chemical_Concentration'] < lower_bound) |  
(df['Chemical_Concentration'] > upper_bound)]
```

- ▶ **Column Renaming:** Renamed complex or technical column headers to make them more readable.

```
df.rename(columns={  
'region_loc_01': 'Region',  
'temp_celsius': 'Temperature',  
'turb_ntu': 'Turbidity'  
, inplace=True)
```

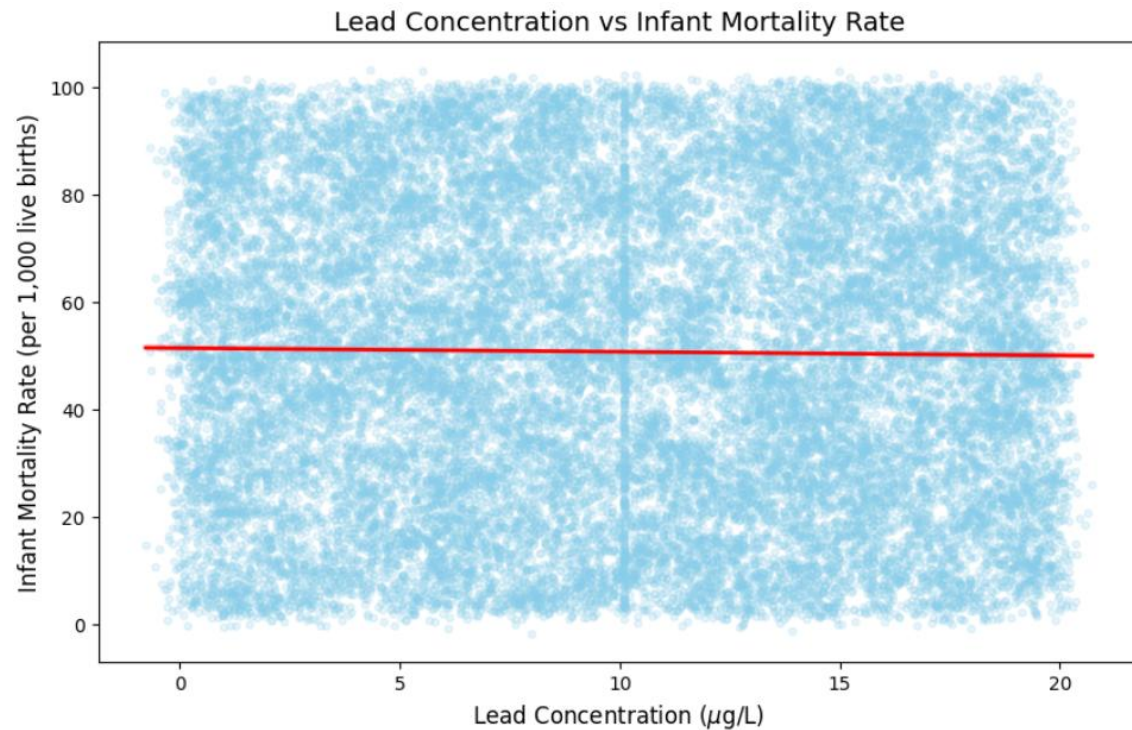
Bar Chart

- ▶ Compares average contamination levels across regions.
- ▶ Identifies specific locations with high pollution spikes.
- ▶ Helps prioritize regional monitoring and intervention.



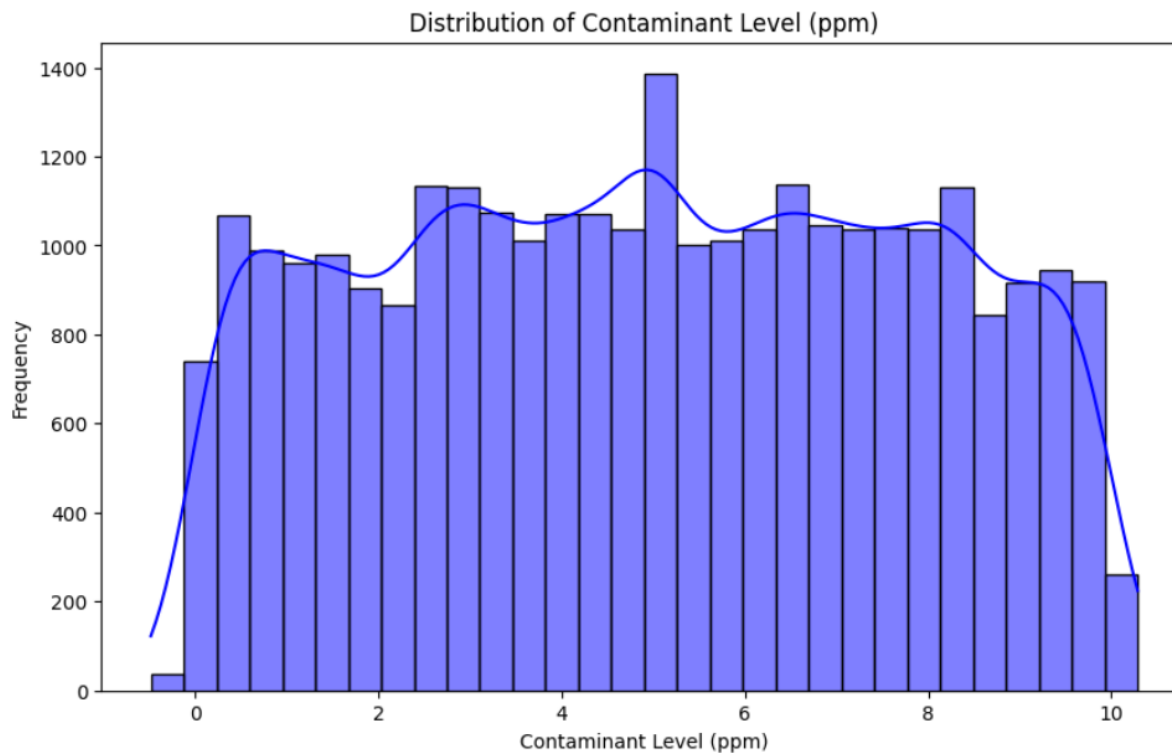
Scatter Plot

- ▶ Shows link between lead levels and infant mortality.
- ▶ Identifies health impacts of chemical concentration.
- ▶ Detects critical trends and outliers in health data.



Histogram

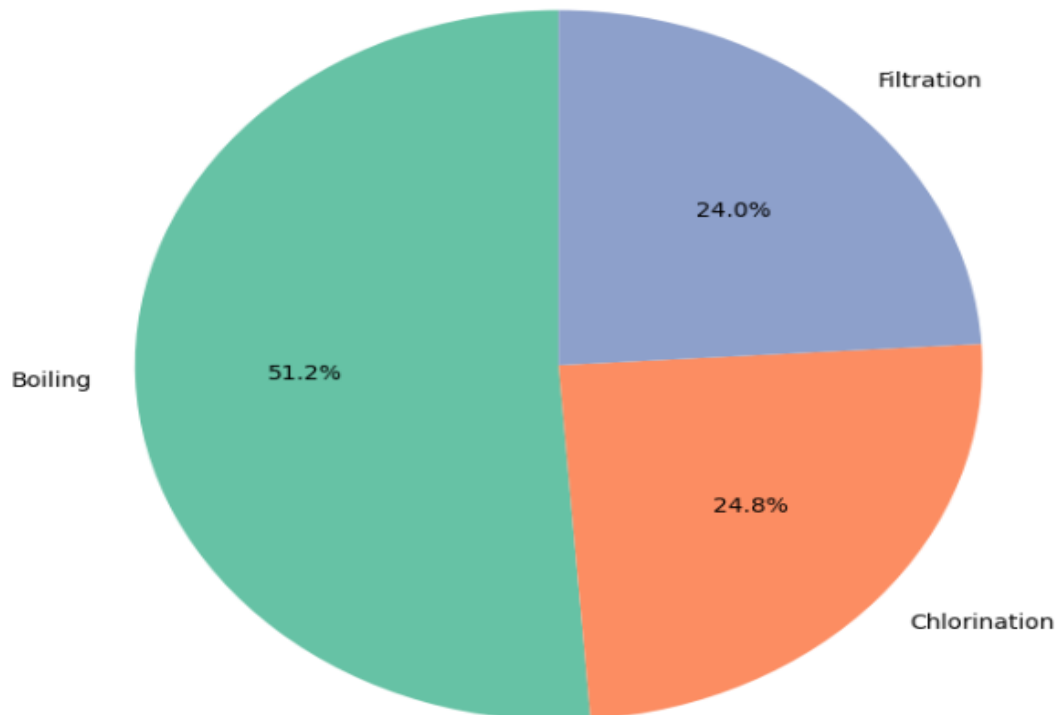
- ▶ Displays frequency of different contaminant levels.
- ▶ Identifies the most common range of pollution.
- ▶ Helps detect skewness or outliers in water safety.



Pie Chart

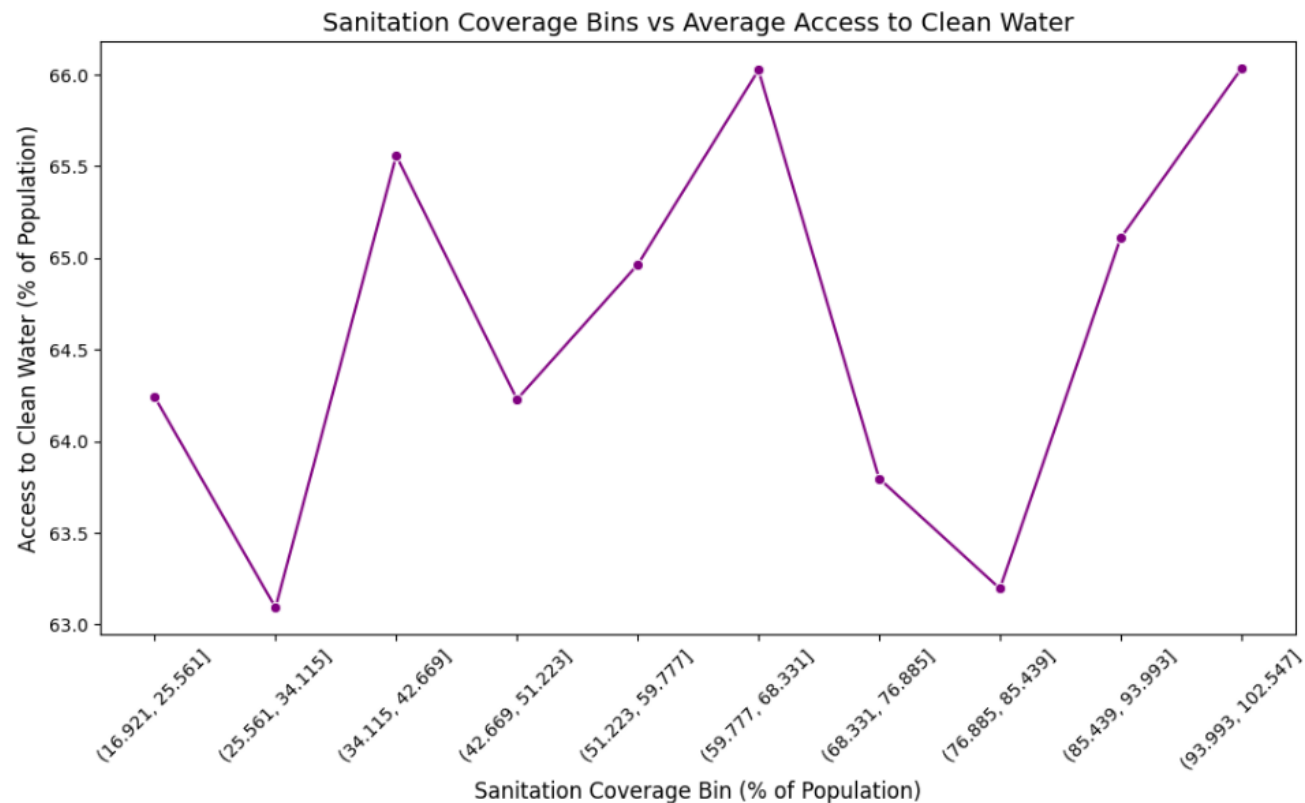
- ▶ Illustrates the market share of different water treatment methods.
- ▶ Highlights Boiling as the most dominant technique at 51.2%.

Most Frequently Used Water Treatment Methods



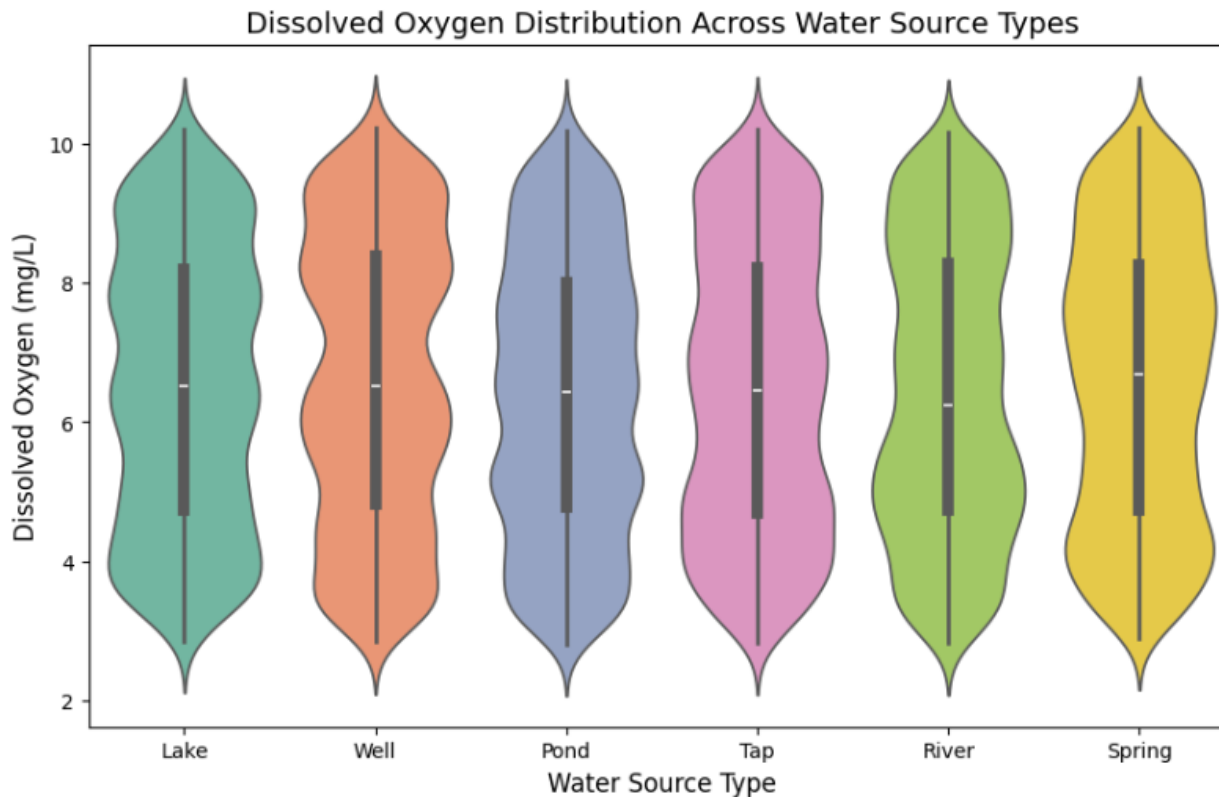
Line Plot

- ▶ Tracks the relationship between sanitation coverage and clean water access.
- ▶ Visualizes fluctuations in water availability across different population bins.



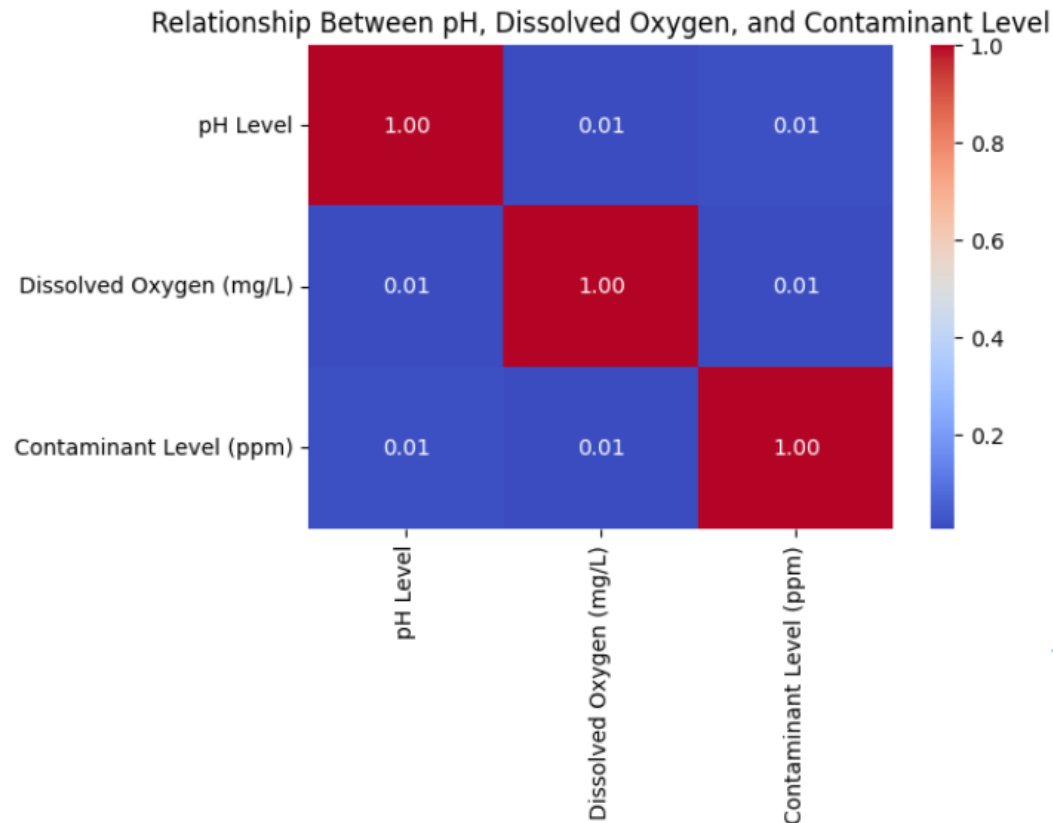
Violin Plot

- ▶ Combines box plot and density traces to show data distribution.
- ▶ Displays the range and concentration of dissolved oxygen across sources.
- ▶ Helps identify variations in water health and oxygen consistency.



Heatmap

- ▶ Visualizes the correlation between pH, Oxygen, and Contaminant levels.
- ▶ Helps identify if certain metrics act as predictors for water pollution.



Key Insights

- ▶ **Policy over Wealth:** Higher GDP doesn't guarantee clean water; strong government policy matters more.
- ▶ **Hidden Bacterial Risk:** Wells are just as contaminated as rivers; natural sources are not "safer."
- ▶ **Deceptive Metrics:** Water can have perfect pH and Oxygen levels while still being highly toxic.
- ▶ **Infrastructure Mismatch:** Sanitation and clean water access grow separately, showing a lack of integrated planning.
- ▶ **No Universal Treatment:** No single "best" method exists; safe water requires combining different treatment types.

Recommendations

- ▶ Prioritize policy and infrastructure over general economic growth.
- ▶ Mandate water treatment for all sources, including wells.
- ▶ Implement multi-metric testing beyond basic pH and oxygen.
- ▶ Integrate sanitation and water supply infrastructure planning.
- ▶ Deploy multi-barrier treatment frameworks to ensure safety.

Conclusion

- ▶ Water pollution is a systemic global issue, not a localized one.
- ▶ Economic wealth alone is insufficient for ensuring water safety.
- ▶ Natural sources like wells require mandatory active treatment.
- ▶ Chemical metrics (pH/Oxygen) do not fully guarantee safe water.
- ▶ Integrated planning of sanitation and supply is critical for health.

The background features abstract, overlapping geometric shapes in various shades of blue, ranging from light sky blue to deep navy blue. These shapes are primarily located on the right side of the frame, creating a dynamic, modern look. The rest of the background is a solid, very light blue.

Thank You